

AI AGENT SAFETY RESEARCH

Literature Survey and Research Gap

Team ALGOX

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Research-stage document based on the team's literature survey.

1. Introduction

AI agents can reason, make decisions, use tools, and perform multi-step tasks. As agent autonomy increases, evaluating safety requires more than checking whether a task was completed. This research reviews four papers to identify major safety dimensions, evaluation approaches, limitations, and the research gap relevant to an AI Agent Safety Evaluation Framework.

2. Research Objectives

- Study existing AI agent safety and evaluation approaches.
- Identify major safety risks and evaluation dimensions.
- Compare enterprise evaluation, safety-layer frameworks, realistic safety testing, and domain-specific risk perspectives.
- Identify limitations and open research areas.
- Define a research direction for a unified AI agent safety evaluation framework.

3. Literature Survey

Paper 1 — AI Agents: Architecture, Safety Frameworks, and Emerging Applications

- **Author:** Fayaz Ali — Southeast European University — 2025.
- **Focus:** Agent architectures, safety risks, controlled experiments, and deployment governance.
- **Safety dimensions:** Alignment, robustness, interpretability, controllability, and societal safety.
- **Architectures:** Reactive, BDI/deliberative, LLM-based, and multi-agent architectures.
- **Four-layer safety framework:** Objective Alignment → Behavioral Constraints → Monitoring & Auditability → Societal Governance.

- **Reported findings:** In 200 benchmark tasks, a Multi-Agent System achieved 89% first-attempt completion and a 70B model with CoT achieved 84%. In adversarial trials, unsafe output was reported for agents, while guardrails reduced 70B unsafe output from 12% to 7%.
- **Advantages:** Connects prevention, runtime controls, monitoring, and governance; addresses prompt injection, goal misspecification, distributional shift, and multi-agent coordination.
- **Open areas:** Stronger prompt-injection defenses, scalable human oversight, longitudinal evaluation under distribution shift, and cross-disciplinary governance.
- **Relevance:** Provides safety controls and governance layers for the proposed evaluation engine.

Paper 2 — Enterprise Agent Evaluation Framework for Autonomous AI Systems (EAEF)

- **Authors/Source:** Baradwaj Bandi Sudakara — IJFIST, Volume 7, Issue 4, July–August 2024.
- **Goal:** Evaluate enterprise AI agents beyond isolated benchmark accuracy.
- **12 dimensions:** Reasoning, groundedness, hallucination, tools, workflow, memory, multi-agent coordination, security/compliance, latency, cost, reliability, and human satisfaction.
- **Scoring:** Weighted Enterprise Agent Score (EAS) with continuous evaluation in CI/CD / DevSecOps.
- **Case study:** Autonomous software quality-engineering agent for enterprise API testing, using workflow execution, API interactions, reasoning traces, memory updates, logs, and user feedback.
- **Reported results:** Overall EAS = 0.92; Security Compliance = 0.97; Hallucination Detection = 0.95; Workflow Completion = 0.94; Reliability = 0.94; Human Satisfaction = 0.92.
- **Advantages:** Holistic technical, operational, governance, and human-centered evaluation; extensible with domain-specific metrics.
- **Open areas:** Adaptive weighting, self-evolving metrics, multimodal agents, federated evaluation, explainable EAS, and enterprise-specific benchmarks.
- **Relevance:** Strong basis for scoring architecture and the overall evaluation lifecycle.

Paper 3 — OPENAGENTS SAFETY: A Comprehensive Framework for Evaluating Real-World AI Agent Safety

- **Authors:** Sanidhya Vijayvargiya, Aditya Bharat Soni, Xuhui Zhou, Zora Zhiruo Wang, Nouha Dziri, Graham Neubig, Maarten Sap.
- **Publication:** ICLR 2026.
- **Scale:** 350+ executable tasks across 8 safety-risk categories.
- **Environment and tools:** Browser, code execution, file system, bash terminal, messaging, and sandboxed Docker environments.
- **Evaluation design:** Multi-turn and multi-user scenarios with benign, ambiguous, and adversarial intents; rule-based evaluator plus LLM-as-Judge.
- **Reported findings:** Seven LLMs showed unsafe behavior in approximately 49%–73% of safety-vulnerable tasks.
- **Advantages:** More realistic than static benchmarks because it uses real tools, long-horizon tasks, social interaction, and adversarial intent; hybrid evaluation catches persistent harmful outcomes and attempted/subtle unsafe behavior.

- **Limitations:** Agents may fail before reaching safety-vulnerable states; dynamic environments and NPC deviations make realistic execution difficult to scale.
- **Relevance:** Closest supplied paper to the project's target; directly informs safety scenarios, risk taxonomy, and evaluator design.

Paper 4 — Advancing Innovation in Financial Stability: A Comprehensive Review of AI Agent Frameworks, Challenges and Applications

- **Author/Source:** Satyadhar Joshi — World Journal of Advanced Engineering Technology and Sciences, 2025, 14(02), 117–126.
- **DOI:** 10.30574/wjaets.2025.14.2.0071.
- **Frameworks reviewed:** LangGraph, CrewAI, AutoGen, and other agent frameworks.
- **Applications:** Trading, investment analysis, risk assessment, fraud detection, and financial decision support.
- **Challenges:** Reliability, explainability, ethics, autonomous-system risk, regulatory compliance, model robustness, transparency, interpretability, security, data quality, and scalability.
- **Advantages:** Broad comparison of agent frameworks and their suitability for complex financial tasks; connects agent technology with risk management and regulatory requirements.
- **Open problems:** Reliability and explainability in financial decisions, regulatory compliance, ethical concerns, data quality, and scalable deployment.
- **Future direction:** More robust and explainable models, better data quality, stronger regulatory/ethical controls, and improved multi-agent systems.
- **Relevance:** Adds domain-specific safety requirements such as financial risk, compliance, explainability, and security.

4. Comparative Research Findings

Research Work	Main Contribution	Use for Our Research
Fayaz Ali (2025)	Layered safety controls and governance	Safety layers, controls, oversight
EAEF (2024)	12-dimensional enterprise evaluation + EAS	Metrics and scoring architecture
OPENAGENTSAFETY	2026)Realistic executable safety tasks	Scenarios, risk taxonomy, evaluator
Joshi (2025)	Framework review and domain-specific risk	s Compliance, explainability, security

5. Key Safety Evaluation Criteria

- Goal / Objective Alignment
- Prompt-injection resistance
- Privacy and security
- Hallucination and groundedness
- Robustness and reliability
- Tool usage and unsafe tool actions
- Workflow completion
- Compliance and governance
- Human oversight and satisfaction
- Multi-agent behavior and coordination
- Domain-specific risk requirements

6. Research Gap

The reviewed papers address different parts of AI agent evaluation. EAEF provides multi-dimensional enterprise quality scoring; Fayaz Ali provides layered safety controls; OPENAGENTSAFETY provides realistic executable safety evaluation; and Joshi highlights domain-specific risks and governance. The key gap identified in the research is the absence, within the supplied literature, of a single unified approach that combines these perspectives into one comprehensive framework.

7. Proposed Research Direction

- Combine multi-dimensional scoring with preventive safety guardrails.
- Use realistic executable safety tasks and multi-turn tool-based scenarios.
- Use hybrid rule-based and LLM-based evaluation.
- Evaluate alignment, prompt-injection resistance, privacy, security, unsafe tool use, reliability,compliance, and human oversight.
- Generate a final safety/readiness score with detailed failure reporting and continuous monitoring.

8. Conclusion

The literature survey shows that AI-agent evaluation must go beyond task accuracy and consider safety, reliability, tool use, governance, and real-world behavior. OPENAGENTSAFETY provides the closest foundation for realistic safety benchmarking, while EAEF provides a strong foundation for overall evaluation and scoring. Together with the layered safety and domain-risk perspectives from the other papers, these studies provide a strong research base for the proposed AI Agent Safety Evaluation Framework.

9. References

1. Fayaz Ali, *AI Agents: Architecture, Safety Frameworks, and Emerging Applications*, 2025.
2. *Enterprise Agent Evaluation Framework for Autonomous AI Systems: A Multi-Dimensional Approach to Reliability, Safety, and Quality Assurance*, 2024.
3. *OPENAGENTSAFETY: A Comprehensive Framework for Evaluating Real-World AI Agent Safety*, ICLR 2026.
4. Satyadhar Joshi, *Advancing Innovation in Financial Stability: A Comprehensive Review of AI Agent Frameworks, Challenges and Applications*, 2025. DOI: 10.30574/wjaets.2025.14.2.0071.